

Software Engineering Lab (SE Lab) — Lab 2 Report

Agile Backlog Creation & Sprint Simulation in Jira

Student Details:

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- **Problem Statement:** #48 — Incident Escalation & On-Call Rotation Engine
- **GitHub Repository:** github.com/arjunsrikanthh/PES1UG24AM920_se_lab
- **Jira Workspace:** arjun-srikanth.atlassian.net (Project: Software-development / SD board)

1. Project Overview & Scenario

The system is an Incident Escalation & On-Call Rotation Engine designed to organize team shift rotations, route incoming alerts from monitoring platforms through multi-tiered escalation ladders (webhooks, SMS, voice, email), and generate incident post-mortems. In this lab, the functional requirements identified in Lab 1 were decomposed into Epics and User Stories, estimated using Planning Poker with the Fibonacci sequence, and tracked across a simulated Scrum sprint using Jira.

2. Product Backlog: Epics and User Stories

The backlog was structured into five distinct Epics covering the end-to-end incident lifecycle.

Epic 1: On-Call Rotation & Schedule Management

- **Description:** Manage recurring rotation schedules, shift intervals, manual overrides, and holiday substitutions.

- **Story 1.1 (SD-6): Recurring Shift Rotations**
 - *User Story:* As an Incident Commander, I want to configure recurring shift schedules with custom durations, so that on-call coverage is maintained 24/7 without manual scheduling gaps.
 - *Story Points:* 5 | *Priority:* Medium | *Status:* In Progress (Rolled over to Sprint 2)
- |---| **Story 1.2 (SD-7): Shift Overrides & Holiday Swaps**
 - *User Story:* As an on-call engineer, I want to request and log shift overrides and holiday swaps, so that temporary coverage changes are automatically applied to the active rotation.
 - *Story Points:* 3 | *Priority:* Medium | *Status:* Done (Sprint 1)

Epic 2: Alert Ingestion & Verification Pipeline

- **Description:** Ingest alerts from monitoring systems via authenticated webhooks and resolve the active on-call engineer.
- **Story 2.1 (SD-8): Webhook Alert Ingestion**
 - *User Story:* As a monitoring system, I want to send alert payloads via authenticated webhooks within 3 seconds, so that incident data is captured securely.
 - *Story Points:* 5 | *Priority:* High | *Status:* Done (Sprint 1)
- **Story 2.2 (SD-9): Active Responder Resolution**
 - *User Story:* As the alerting engine, I want to query the rotation schedule to resolve the active primary engineer for the affected service, so that alerts reach the right person immediately.
 - *Story Points:* 3 | *Priority:* High | *Status:* Done (Sprint 1)

Epic 3: Notification & Escalation Engine

- **Description:** Route alerts across multi-tier policies and trigger automated escalation timeouts.
- **Story 3.1 (SD-10): Escalation Ladder Configuration**
 - *User Story:* As an Incident Commander, I want to define multi-tiered escalation ladders and per-tier timeouts, so that alert routing matches operational team policies.
 - *Story Points:* 5 | *Priority:* High | *Status:* To Do (Backlog)
- **Story 3.2 (SD-11): 5-Minute Escalation Trigger**

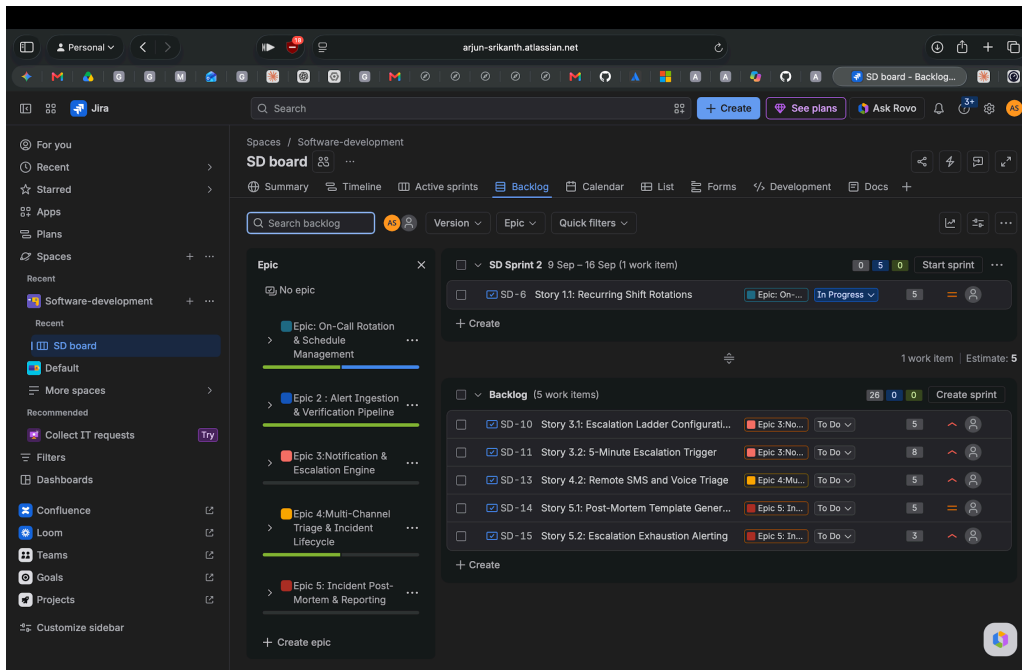
- *User Story:* As the alerting engine, I want to escalate to the secondary engineer if Tier-1 does not acknowledge within 5 minutes, so that critical incidents are never left unattended.
- *Story Points:* 8 | *Priority:* High | *Status:* To Do (Backlog)

Epic 4: Multi-Channel Triage & Incident Lifecycle

- **Description:** Enable multi-channel acknowledgment, reassignment, and resolution of active incidents.
- **Story 4.1 (SD-12): Web Dashboard Triage**
 - *User Story:* As an on-call engineer, I want to view, acknowledge, reassign, and resolve incidents from a web dashboard, so that incident status is updated within 10 seconds.
 - *Story Points:* 5 | *Priority:* High | *Status:* Done (Sprint 1)
- **Story 4.2 (SD-13): Remote SMS and Voice Triage**
 - *User Story:* As an on-call engineer, I want to acknowledge incidents via SMS reply or phone keypress, so that I can triage alerts away from my workstation.
 - *Story Points:* 5 | *Priority:* High | *Status:* To Do (Backlog)

Epic 5: Incident Post-Mortem & Reporting

- **Description:** Support post-incident investigation, template generation, and exhaustion alerting.
- **Story 5.1 (SD-14): Post-Mortem Template Generation**
 - *User Story:* As an Incident Commander, I want the system to auto-generate a post-mortem report template linked to the incident with timeline and root-cause fields, so that the team can conduct structured reviews.
 - *Story Points:* 5 | *Priority:* Medium | *Status:* To Do (Backlog)
- **Story 5.2 (SD-15): Escalation Exhaustion Alerting**
 - *User Story:* As an Incident Commander, I want to receive immediate alerts when all escalation tiers are exhausted, so that critical unhandled incidents receive urgent manual intervention.
 - *Story Points:* 3 | *Priority:* Medium | *Status:* To Do (Backlog)



3. Story Point Estimation & Planning Poker Rationale

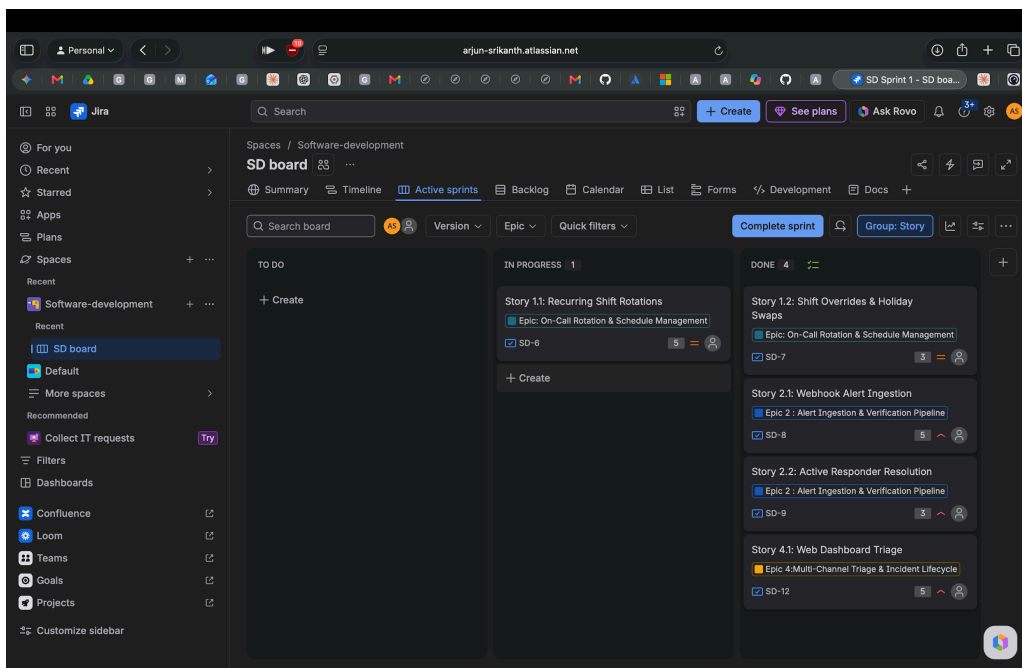
Story points were estimated using the Fibonacci scale (1, 2, 3, 5, 8) based on complexity, effort, and technical uncertainty:

- 3 Points (SD-7, SD-9, SD-15): Well-defined functionality with low technical uncertainty, taking approximately half a day to one day of effort.
- 5 Points (SD-6, SD-8, SD-10, SD-12, SD-13, SD-14): Standard features involving moderate logic, external API integrations, or state updates, estimated at 1.5 to 2 days.
- 8 Points (SD-11): High-complexity asynchronous workflow involving timer scheduling, race condition handling, and multi-tier state management.

4. Sprint 1 Simulation & Execution

- Sprint Name: SD Sprint 1
- Duration: 1 week (simulated on 02/Sep/2026)
- Initial Commitment: 5 User Stories totaling 21 Story Points (SD-6, SD-7, SD-8, SD-9, SD-12).

- Completed Work (16 Story Points):
 - SD-7: Shift Overrides & Holiday Swaps (3 pts) — Moved from To Do to In Progress to Done.
 - SD-8: Webhook Alert Ingestion (5 pts) — Moved from To Do to In Progress to Done.
 - SD-9: Active Responder Resolution (3 pts) — Moved from To Do to In Progress to Done.
 - SD-12: Web Dashboard Triage (5 pts) — Moved from To Do to In Progress to Done.
- Incomplete Work (5 Story Points):
 - SD-6: Story 1.1: Recurring Shift Rotations (5 pts) — Remained in In Progress at sprint closure and rolled over to SD Sprint 2 (9 Sep – 16 Sep).
- Velocity Delivered: 16 Story Points.

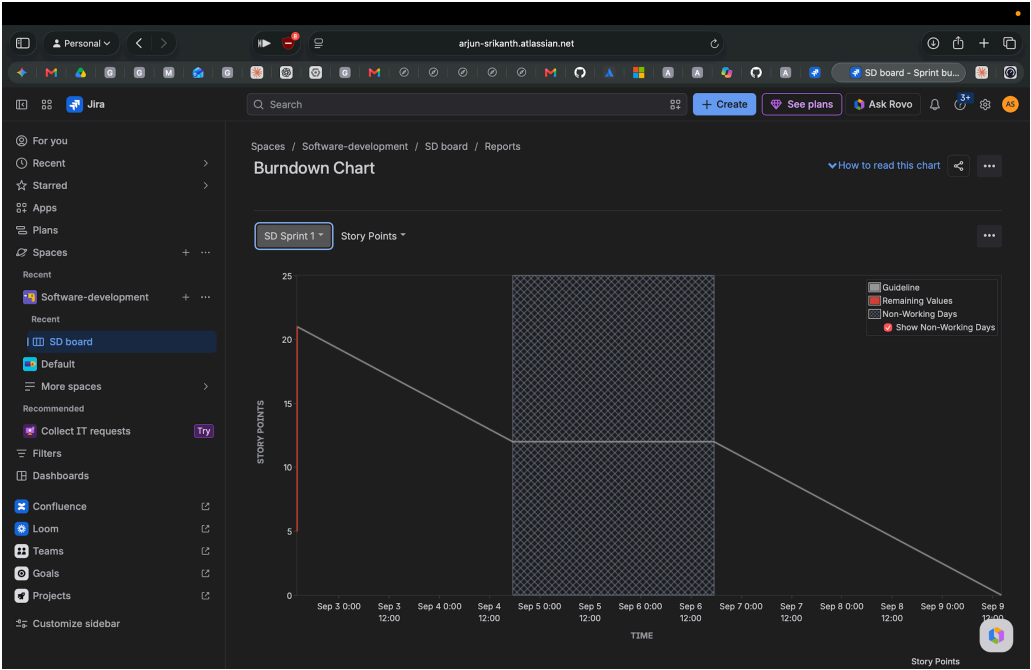


5. Burndown Chart Analysis

The burndown chart tracks story point completion over time:

- Initial sprint commitment was 21 story points at sprint start (2:02 PM).
- 16 story points were burned down as SD-7, SD-8, SD-9, and SD-12 reached Done (2:03 PM).

- 5 story points (SD-6) remained uncompleted at sprint completion (2:05 PM).
- The chart clearly reflects the delta between the guideline trajectory and the remaining work, demonstrating that 76.2% of committed points were successfully burned down.



Date	Issue	Event Type	Event Detail	Inc.	Dec.	Remaining
02/Sep/26 2:02 PM	SD-12	Sprint start		5		
	SD-6			5		
	SD-7			3		
	SD-8			5		
	SD-9			3		
	Total					21
02/Sep/26 2:03 PM	SD-7	Burndown	Issue completed	3		18
	SD-8	Burndown	Issue completed	5		13
	SD-9	Burndown	Issue completed	3		10
	SD-12	Burndown	Issue completed	5		5
02/Sep/26 2:05 PM	SD-12	Sprint ended by Arjun Srikanth				0
	SD-6					5
	SD-7					0
	SD-8					0
	SD-9					0
	Total					5

6. Answers to Reflection Questions

1. Did your estimations reflect the actual effort?

The Fibonacci estimates accurately reflected the relative differences in complexity across most stories. Straightforward tasks like responder lookup (SD-9, 3 points) and shift overrides (SD-7, 3 points) were completed smoothly. However, SD-6 (Recurring Shift Rotations, 5 points) proved more intricate due to calendar edge cases and recurrence math, preventing its completion within the sprint. This demonstrated that recurring schedule logic carried higher uncertainty and should have been estimated at 8 points or decomposed into smaller stories.

2. Was your backlog well-prioritized?

Yes. The backlog was prioritized by functional dependencies and operational risk. Ingestion (Epic 2) and rotation scheduling (Epic 1) formed the foundation required for alert dispatch. Web triage (SD-12) was included to provide immediate visibility. Dependent items such as multi-tier escalation triggers (Epic 3), voice/SMS remote triage (Epic 4), and post-mortems (Epic 5) were appropriately scheduled for subsequent sprints.

3. How did your simulated sprint align with your plan?

The team planned 21 story points and delivered 16 story points (4 out of 5 stories completed, representing 76.2% of the planned scope). While the goal was full completion, the simulation realistically demonstrated how Scrum manages incomplete work: SD-6 was cleanly carried over into Sprint 2 without compromising quality or rushing the release.

4. What insights did the burndown chart give about your team's capacity?

The burndown chart showed that the team's sustainable velocity for a 1-week sprint is approximately 16 story points. Committing to 21 points exceeded team capacity. For future sprint planning (such as SD Sprint 2), 15–16 story points should serve as the planning baseline. Furthermore, the sharp drops in the burndown highlight that tasks were completed in batches during the simulation,

reinforcing the value of continuous daily progress tracking in actual development cycles.